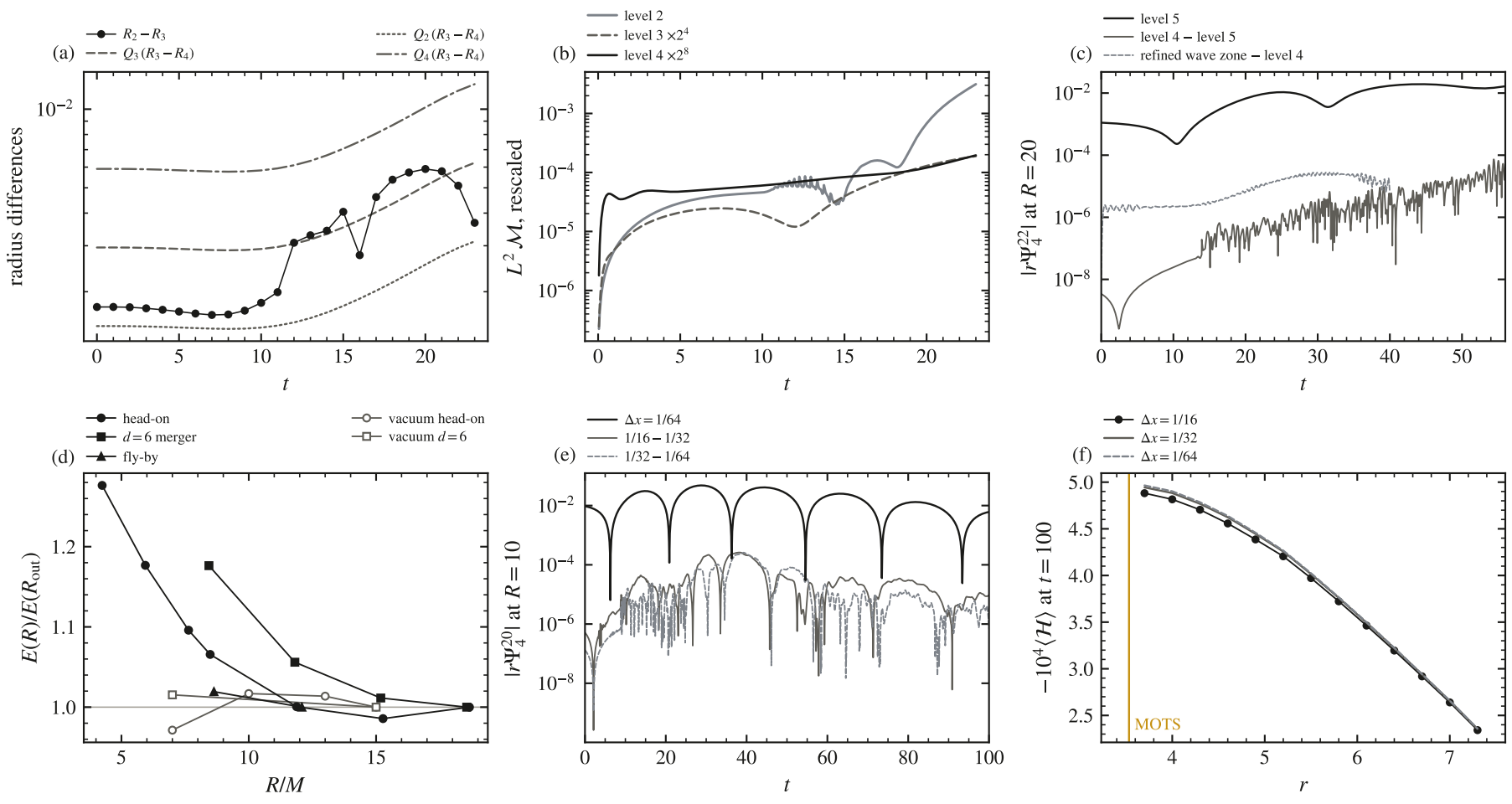

(a)
$R_2 - R_3$
$Q_3\,(R_3 - R_4)$
$Q_2\,(R_3 - R_4)$
$Q_4\,(R_3 - R_4)$
radius differences
10^{-2}
0
5
10
15
20
t
(b)
level 2
level 3 $\times 2^4$
level 4 $\times 2^8$
$L^2\,\mathcal{M}$, rescaled
10^{-3}
10^{-4}
10^{-5}
10^{-6}
0
5
10
15
20
t
(c)
level 5
level 4 – level 5
refined wave zone – level 4
$|r\Psi_4^{22}|$ at $R = 20$
10^{-2}
10^{-4}
10^{-6}
10^{-8}
0
10
20
30
40
50
t
(d)
head-on
$d = 6$ merger
fly-by
vacuum head-on
vacuum $d = 6$
$E(R)/E(R_{\rm out})$
1.2
1.1
1.0
5
10
15
R/M
(e)
$\Delta x = 1/64$
1/16 – 1/32
1/32 – 1/64
$|r\Psi_4^{20}|$ at $R = 10$
10^{-2}
10^{-4}
10^{-6}
10^{-8}
0
20
40
60
80
100
t
(f)
$\Delta x = 1/16$
$\Delta x = 1/32$
$\Delta x = 1/64$
$-10^4\langle\mathcal{H}\rangle$ at $t = 100$
5.0
4.5
4.0
3.5
3.0
2.5
MOTS
4
5
6
7
r